

Experiment No.1

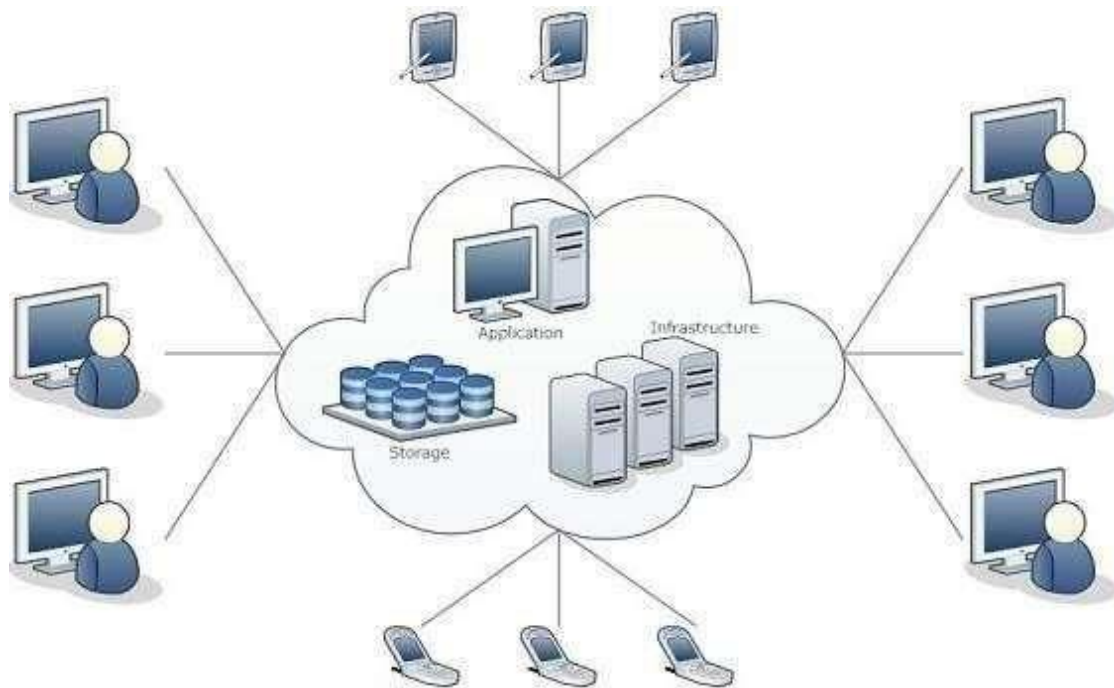
Aim: Study of NIST Model of Cloud Computing.

Theory:

Cloud computing is on-demand access, via the internet, to computing resources—applications, servers (physical servers and virtual servers), data storage, development tools, networking capabilities, and more—hosted at a remote data center managed by a cloud services provider (or CSP). The CSP makes these resources available for a monthly subscription fee or bills them according to usage.

Overview:

Cloud Computing refers to manipulating, configuring, and accessing the hardware and software resources remotely. It offers online data storage, infrastructure, and application. Cloud computing offers platform independency, as the software is not required to be installed locally on the PC. Hence, the Cloud Computing is making our business applications mobile and collaborative.



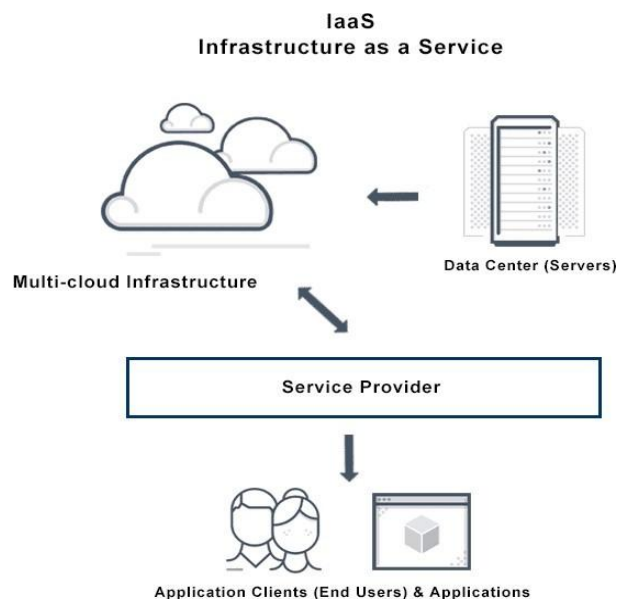
Types Of Services:

There are the following three types of cloud service models -

1. Infrastructure as a Service (IaaS)
2. Platform as a Service (PaaS)
3. Software as a Service (SaaS)

1. Infrastructure as a Service (IaaS):

IaaS is also known as Hardware as a Service (HaaS). It is a computing infrastructure managed over the internet. The main advantage of using IaaS is that it helps users to avoid the cost and complexity of purchasing and managing the physical servers. This is the most basic level of cloud service. Think of it as renting the building blocks of IT infrastructure, like virtual servers, storage space, and networking equipment. You have complete control over these resources and can configure them to your specific needs, just like you would with your own physical hardware. Popular IaaS providers include Amazon Web Services (AWS), Microsoft Azure, and Google Cloud Platform.



Characteristics of IaaS:

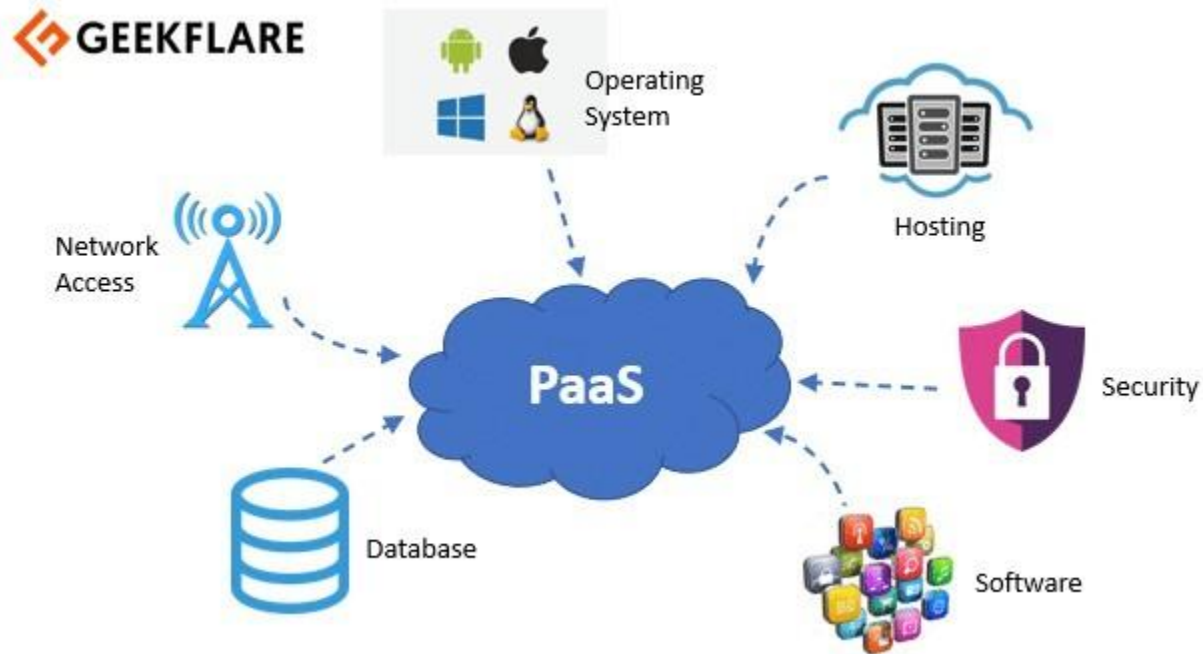
There are the following characteristics of IaaS –

- Resources are available as a service
- Services are highly scalable
- Dynamic and flexible
- GUI and API-based access
- Automated administrative tasks.

Examples: AWS EC2, Microsoft Azure Virtual Machines, Google Compute Engine, IBM Cloud Virtual Servers, Oracle Cloud Infrastructure Compute, Alibaba Cloud ECS, DigitalOcean Droplets, Linode Compute, Vulture Cloud Compute, Rackspace Cloud Servers.

2. Platform as a Service (PaaS):

This is a step up from IaaS, providing a complete development and deployment environment in the cloud. Think of it as getting a pre-built apartment with all the essential utilities and furniture already in place. You don't need to worry about managing the underlying infrastructure; you can focus on developing and running your applications on the platform.



Characteristics of PaaS:

There are the following characteristics of PaaS -

- Accessible to various users via the same development application.
- Integrates with web services and databases.
- Builds on virtualization technology, so resources can easily be scaled up or down as per the organization's need.
- Support multiple languages and frameworks.
- Provides an ability to "Auto-scale".

Example: AWS Elastic Beanstalk, Windows Azure, Heroku, Force.com, Google App Engine, Apache Stratos, Magento Commerce Cloud, and OpenShift.

3. Software as a Service (SaaS):

This is the most user-friendly level of cloud service. Think of it as renting ready-to-use applications, like email, office productivity tools, or customer relationship management (CRM) software. You simply access the software through a web browser or mobile app, without the need for installation or maintenance. Popular SaaS providers include Microsoft Office 365, Google Workspace, and Salesforce.



Characteristics of SaaS:

There are the following characteristics of SaaS -

- Managed from a central location
- Hosted on a remote server
- Accessible over the internet
- Users are not responsible for hardware and software updates. Updates are applied automatically.
- The services are purchased on the pay-as-per-use basis

Example: Big Commerce, Google Apps, Sales force, Dropbox, ZenDesk, Cisco WebEx, ZenDesk, Slack, andGo to Meeting.

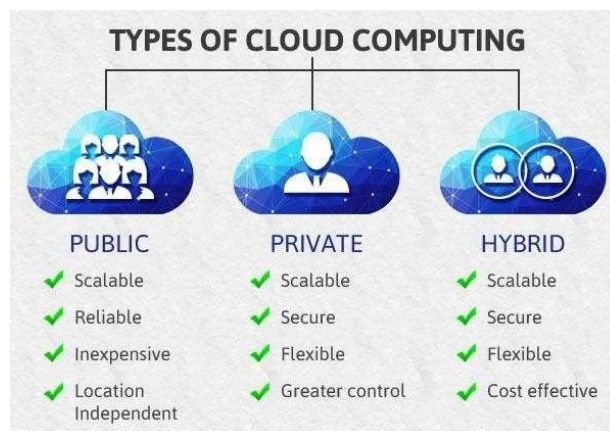
Types Of Cloud Computing

The way cloud resources are delivered can be categorized into three main types:

1. Public Cloud:

This is the most common type of cloud, where resources are shared among multiple users from different organizations. Think of it as a large apartment building where many tenants share the facilities. Public clouds offer the benefits of scalability and cost-effectiveness, but may raise concerns about security and data privacy. Popular public cloud providers include AWS, Azure, and Google Cloud Platform.

Public clouds are managed by third parties that provide cloud services over the internet to the public, these services are available as pay-as-you-go billing models. They offer solutions for minimizing IT infrastructure costs and become a good option for handling peak loads on the local infrastructure. Public clouds are the go-to option for small enterprises, which can start their businesses without large upfront investments by completely relying on public infrastructure for their IT needs. The fundamental characteristics of public clouds are multitenancy. A public cloud is meant to serve multiple users, not a single customer. A user requires a virtual computing environment that is separated, and most likely isolated, from other users.



Advantages of using a public cloud are:

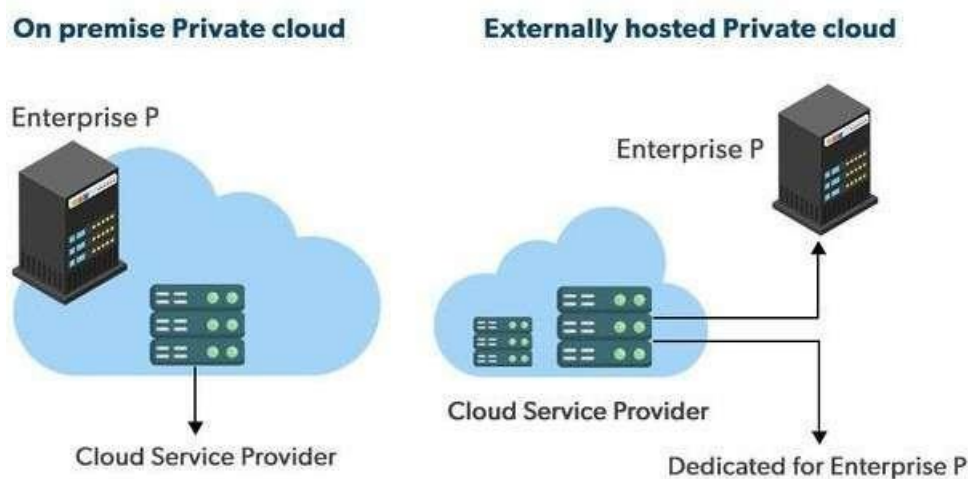
- High Scalability
- Cost Reduction
- Reliability and flexibility
- Disaster Recovery

Disadvantages of using a public cloud are:

- Loss of control over data
- Data security and privacy
- Limited Visibility
- Unpredictable cost

2. Private cloud

Private clouds are distributed systems that work on private infrastructure and provide the users with dynamic provisioning of computing resources. Instead of a pay-as-you-go model in private clouds, there could be other schemes that manage the usage of the cloud and proportionally billing of the different departments or sections of an enterprise. Private cloud providers are HP Data Centers, Ubuntu, Elastic-Private cloud, Microsoft, etc. This is a dedicated cloud environment used by a single organization. Think of it as having your own private apartment building with complete control and security. Private clouds offer higher levels of security and customization, but can be more expensive and require more technical expertise to manage. Some organizations build their own private clouds, while others use private cloud services from providers like VMware or OpenStack.



The advantages of using a private cloud are as follows:

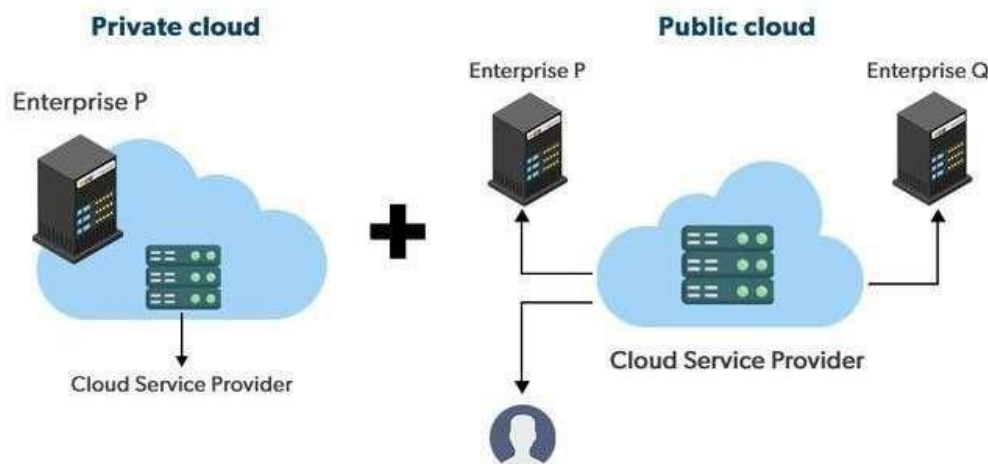
- Customer information protection: In the private cloud security concerns are less since customer data and other sensitive information do not flow out of private infrastructure.
- Infrastructure ensuring SLAs: Private cloud provides specific operations such as appropriate clustering, data replication, system monitoring, and maintenance, disaster recovery, and other uptime services.
- Compliance with standard procedures and operations: Specific procedures have to be put in place when deploying and executing applications according to third-party compliance standards. This is not possible in the case of the public cloud.

Disadvantages of using a private cloud are:

- The restricted area of operations: Private cloud is accessible within a particular area. So, the area of accessibility is restricted.
- Expertise requires: In the private cloud security concerns are less since customer data and other sensitive information do not flow out of private infrastructure. Hence skilled people are required to manage & operate cloud services.

3. Hybrid cloud:

A hybrid cloud is a heterogeneous distributed system formed by combining facilities of the public cloud and private cloud. For this reason, they are also called heterogeneous clouds. A major drawback of private deployments is the inability to scale on-demand and efficiently address peak loads. Here public clouds are needed. Hence, a hybrid cloud takes advantage of both public and private clouds. This is a combination of public and private clouds, allowing organizations to leverage the benefits of both. Think of it as having two apartments, one for everyday use and the other for more sensitive needs. Hybrid clouds offer flexibility and agility, enabling organizations to move workloads between public and private clouds based on their requirements. Many organizations are adopting hybrid cloud strategies to optimize their IT infrastructure and costs.



Advantages of using a Hybrid cloud are:

- Cost: Available at a cheap cost than other clouds because it is formed by a distributed system.
- Speed: It is efficiently fast with lower cost; it reduces the latency of the data transfer process.
- Security: Most important thing is security. A hybrid cloud is totally safe and secure because it works on the distributed system network.

Disadvantages of using a Hybrid cloud are:

- It's possible that businesses lack the internal knowledge necessary to create such a hybrid environment. Managing security may also be more challenging. Different access levels and security considerations may apply in each environment.
- Managing a hybrid cloud may be more difficult. With all of the alternatives and choices available today, not to mention the new PaaS components and technologies that will be released every day going forward, public cloud and migration to public cloud are already complicated enough. It could just feel like a step too far to include hybrid.

Applications of Cloud Computing in real-world:

Cloud Service Providers (CSP) are providing many types of cloud services and now if we will cloud computing has touched every sector by providing various cloud applications. Sharing and managing resources are easy in cloud computing that's why it is one of the dominant fields of computing. These properties have made it an active component in many fields. Now let's know some of the real-world applications of cloud computing.

Online Data Storage: Cloud computing allows storing data like files, images, audios, and videos, etc. on the cloud storage. The organization need not set physical storage systems to store a huge volume of business data which costs so high nowadays. As they are growing technologically, data generation is also growing with respect to time, and storing that becoming problem. In that situation, Cloud storage is providing this service to store and access data any time as per requirement.

Backup and Recovery: Cloud vendors provide security from their side by storing safe to the data as well as providing a backup facility to the data. They offer various recovery application for retrieving the lost data. In the traditional way backup of data is a very complex problem and also it is very difficult sometimes impossible to recover the lost data. But cloud computing has made backup and recovery applications very easy where there is no fear of running out of backup media or loss of data.

Big data Analysis: We know the volume of big data is so high were storing that in traditional data management system for an organization is impossible. But cloud computing has resolved that problem by allowing the organizations to store their large volume of data in cloud storage without worrying about physical storage. Next comes analyzing the raw data and finding out insights or useful information from it is a big challenge as it requires high-quality tools for data analytics.

Testing and development: Setting up the platform for development and finally performing different types of testing to check the readiness of the product before delivery requires different types of IT resources and infrastructure. But Cloud computing provides the easiest approach for development as well as testing even if deployment by using their IT resources with minimal expenses. Organizations find it more helpful as they got scalable and flexible cloud services for product development, testing, and deployment.

E-commerce Application: Cloud-based e-commerce allows responding quickly to the opportunities which are emerging. Users respond to the market opportunities as well as the traditional e-commerce responds to the challenges quickly. Cloud-based e-commerce gives a new approach to doing business with minimum amount as well as minimum time possible. Customer data, product data, and other operational systems are managed in cloud environments.

Cloud Computing in Education: Cloud computing in the education sector brings an unbelievable change in learning by providing e-learning, online distance learning platforms, and students information portals to the students. It is a new trend in education that provides an attractive environment for learning, teaching, experimenting, etc. to students, faculty members, and researchers. Everyone associated with field can connect to the cloud of their organization and access data and information from there.

Anti-Virus Applications: Previously, organizations were installing antivirus software within their system even if we will see we personally also keep antivirus software in our system for safety from outside cyber threats. But nowadays cloud computing provides cloud antivirus software which means the software stored in the cloud and monitors your system/organization's system remotely. This antivirus software identifies the security risks and fixes them. Sometimes also they give a feature to download the software.

Conclusion: Thus, we have successful studied about the NIST Model of Cloud Computing.